

Time-independent Schrodinger equation

Consider the Schrodinger equation

$$i\hbar \frac{\partial}{\partial t} \Psi(x, t) = H \Psi(x, t)$$

Let $\Psi(x, t)$ be the wavefunction

$$\Psi(x, t) = \sum_k \xi_k(t) \psi_k(x)$$

where

$$\xi_k(t) = \exp\left(-\frac{i}{\hbar} E_k t\right)$$

Then for all k

$$i\hbar \frac{d}{dt} \xi_k(t) \psi_k(x) = E_k \xi_k(t) \psi_k(x)$$

and by the Schrodinger equation we have

$$E_k \xi_k(t) \psi_k(x) = H [\xi_k(t) \psi_k(x)]$$

If H is independent of t then the right-hand can be factored as

$$E_k \xi_k(t) \psi_k(x) = \xi_k(t) H \psi_k(x)$$

The $\xi_k(t)$ cancel leaving the time-independent Schrodinger equation

$$E_k \psi_k(x) = H \psi_k(x)$$

By linearity of H we also have

$$\sum_k E_k \psi_k(x) = H \sum_k \psi_k(x)$$

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